

Task 2: Exploratory Data Analysis (EDA)

Introduction

Exploratory Data Analysis (EDA) is an important step in the data science and machine learning workflow. It involves examining and visualizing data to understand its structure, identify patterns, detect anomalies, and gain insights before applying machine learning algorithms.

Objective

The main objective of this task is to analyze the dataset and understand its characteristics through statistical summaries and visualizations. This helps in making informed decisions during data preprocessing and model development.

Tools and Libraries Used

- Python
- Pandas
- Matplotlib
- Seaborn

Steps Performed

1. Data Loading

The dataset was loaded into a Pandas DataFrame for analysis and manipulation.

2. Data Inspection

The structure of the dataset was examined by:

- Viewing sample records
- Checking the number of rows and columns
- Identifying data types of each feature

3. Missing Value Analysis

The dataset was checked for missing or null values to ensure data quality and determine if preprocessing was required.

4. Statistical Summary

Descriptive statistics such as:

- Mean
- Median
- Standard Deviation
- Minimum Value
- Maximum Value

were calculated to understand the distribution of data.

5. Distribution Analysis

Histograms were used to visualize the distribution of numerical features and identify:

- Data spread
- Skewness
- Unusual patterns

6. Correlation Analysis

A correlation heatmap was generated to examine relationships between variables and identify strongly correlated features.

7. Outlier Detection

Boxplots were used to detect outliers and extreme values that could affect model performance.

8. Pattern Identification

The data was analyzed to identify:

- Trends
- Relationships between variables
- Potential anomalies
- Feature behavior

Visualizations Used

Histogram

Purpose:

- Understand feature distributions
- Detect skewness
- Analyze data spread

Correlation Heatmap

Purpose:

- Identify relationships between variables
- Find highly correlated features

Boxplot

Purpose:

- Detect outliers
- Identify extreme observations

Results and Observations

- Dataset structure was successfully analyzed.
- Missing values were identified and reviewed.
- Feature distributions were examined using histograms.
- Correlations between variables were visualized through a heatmap.

- Outliers and anomalies were detected using boxplots.
- Important patterns and trends were identified within the dataset.

```
import pandas as pd
```

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
# Sample dataset
```

```
data = {
```

```
    'Age': [22, 25, 47, 52, 46, 56, 48, 55, 60, 61],
```

```
    'Salary': [25000, 30000, 45000, 50000, 48000,  
60000, 52000, 65000, 70000, 72000],
```

```
    'Experience': [1, 2, 10, 12, 11, 15, 13, 16, 18, 20]
```

```
}
```

```
df = pd.DataFrame(data)
```

```
# Display dataset
```

```
print(df)
```

```
# Check for missing values
```

```
print("\nMissing Values:")
```

```
print(df.isnull().sum())
```

```
# Check skewness
```

```
print("\nSkewness:")
```

```
print(df.skew())
```

```
# Distribution Plot
```

```
sns.histplot(df['Salary'], kde=True)
```

```
plt.title('Salary Distribution')
```

```
plt.show()
```

```
# Correlation Heatmap
```

```
sns.heatmap(df.corr(), annot=True,
```

```
cmap='coolwarm')
```

```
plt.title('Correlation Heatmap')
```

```
plt.show()
```

```
# Outlier Detection
```

```
sns.boxplot(x=df['Salary'])
```

```
plt.title('Salary Outliers')
```

```
plt.show()
```

```
print("EDA Completed Successfully!")
```

OUTPUT:

```
...   Age  Salary  Experience
0    22   25000           1
1    25   30000           2
2    47   45000          10
3    52   50000          12
4    46   48000          11
5    56   60000          15
6    48   52000          13
7    55   65000          16
8    60   70000          18
9    61   72000          20
```

```
Missing Values:
```

```
Age           0
```

```
Salary        0
```

```
Experience     0
```

```
dtype: int64
```

```
Skewness:
```

```
Age          -1.177064
```

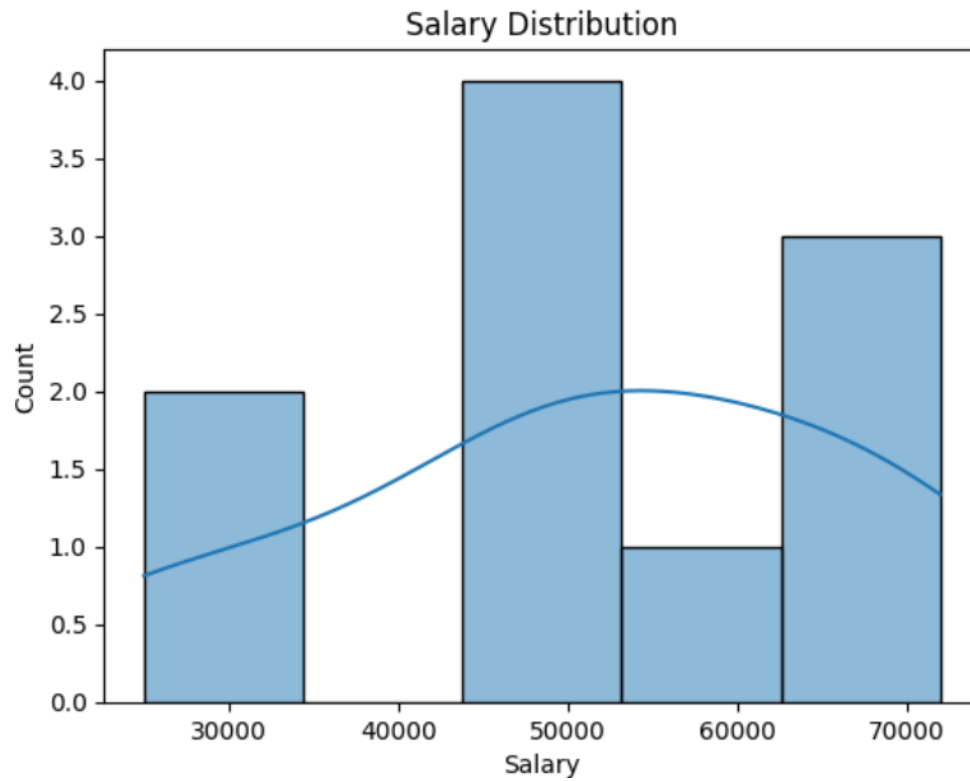
```
Salary       -0.431749
```

```
Experience   -0.744534
```

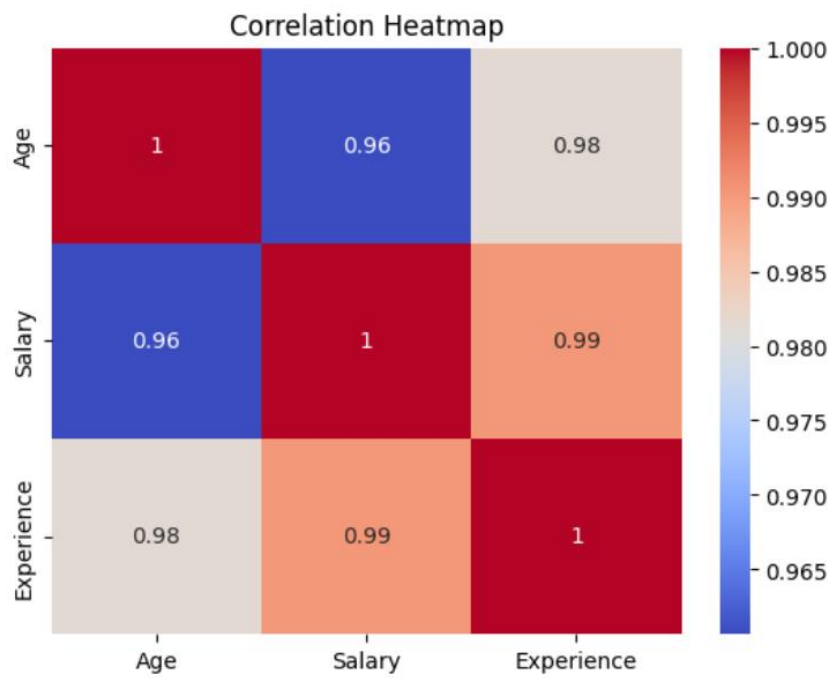
```
dtype: float64
```

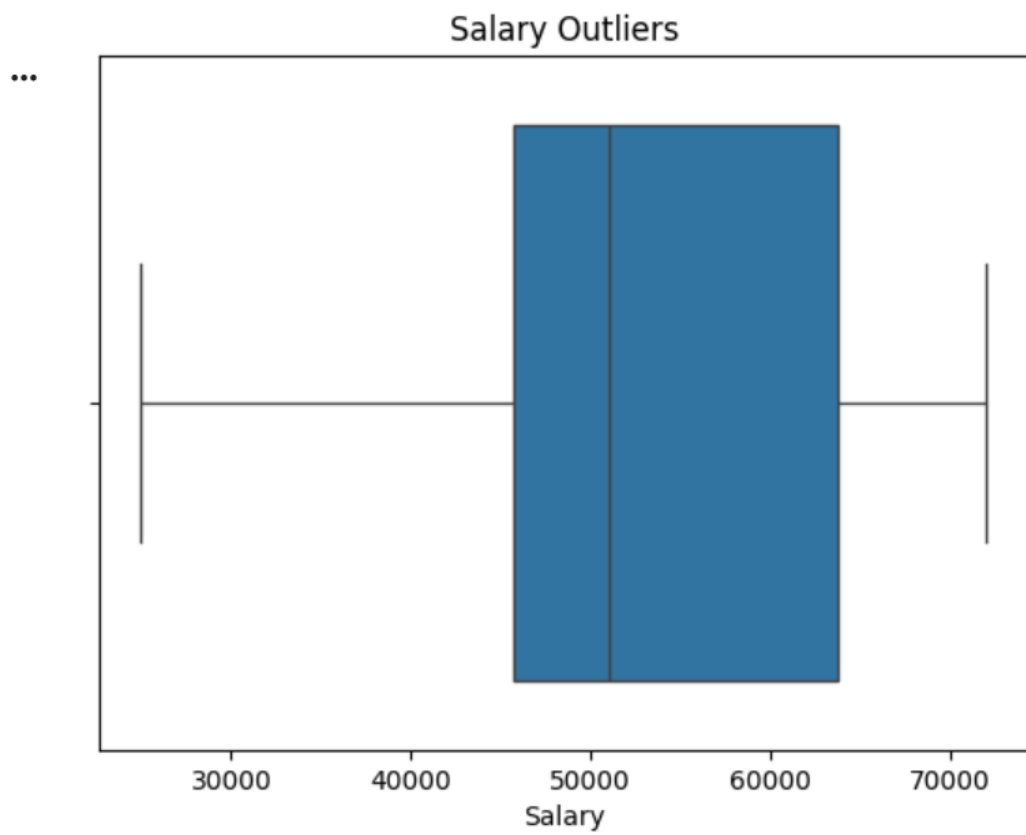
dtype: float64

...



...





EDA Completed Successfully!